

Date:	14-Mar-16	Made by:	VRG
Project No.:	16388377	Checked by:	AT
Subject:	Wexford County Landfill	Reviewed by:	VRG

Project Short Title: GW Discharge Rule 2210(y) Permit Request - Flow Estimate

1.0 Objective

Determine the anticipated daily flow rate and anticipated flow duration from the sump discharge system to the groundwater discharge location.

2.0 Assumptions

Acreage of Lagoon and Spray Curtain is:	4.5	acres
Existing pump is rated at:	35	gallons per minute
Quantity of wastewater to be discharged is:	900,000	gallons
Typical precipitation is:	March	2.0 inches
	April	2.9 inches
	May	3.3 inches

3.0 Calculations

To be conservative, assume that all the precipitation that occurs within the footprint of the lagoon and spray curtain will be captured by the lagoon (no evaporation will occur).

$$\text{Days required for discharge} = \frac{V_L + P_{\text{month}}}{Q_P}$$

Where: V_L = Volume of lagoon
 P_{month} = Precipitation by month
 Q_P = Flow rate of the pump

APRIL 30 days (note: requires discharging precipitation in March as well)

If discharge does not occur until May, then:

MAY 38 days (note: requires discharging precipitation in March and April)

4.0 Reference

<http://www.currentresults.com/Weather/Michigan/precipitation-march.php>